

LECTURE NOTES MATH255

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1. LECTURE 9

In this class we introduce ideals.

1.1. Definition of an ideal. Let R be a ring. In the context of groups, we introduced normal subgroups as the right notion of subgroup to obtain quotient groups. A natural question is: What is the corresponding notion of subring $S \subset R$ that will allow us to obtain a quotient ring R/S ? It turns out that the answer is ideals.

Definition 1.1. Let R be a ring. An additive subgroup $I \subset R$ is an ideal if:

$$(1.1) \quad aI \subset I, \quad Ia \subset I$$

for all $a \in R$, where:

$$(1.2) \quad aI = \{ax \mid x \in I\}, \quad Ia = \{xa \mid x \in I\}.$$

Given a ring R and an ideal I , we can define the quotient R/I as:

$$(1.3) \quad R/I = \{x + I \mid x \in R\}.$$

We claim that:

$$(1.4) \quad (x + I)(y + I) = xy + I$$

is well-defined on R/I . Indeed, if $x' = x + i$, $y' = y + j$ with $i, j \in I$ then:

$$(1.5) \quad x'y' + I = (x + i)(y + j) + I = xy + iy + xj + ij + I = xy + I$$

where the last equality uses $iy, xj, ij \in I$ since $i, j \in I$ and I is an ideal. It follows that $(x + I)(y + I) = xy + I$ is a well-defined binary operation on R/I . We also define $(x + I) + (y + I) = (x + y) + I$, which is well-defined since it's the group operation in the quotient group. Notice that $(R/I, +)$ is an abelian group (since it's the quotient group of an abelian group), and that the product in R/I is associative and satisfies the distributive law (since the corresponding properties hold in R). That is, $(R/I, +, \cdot)$ form a ring. We also obtain a quotient map $\gamma : R \rightarrow R/I$ which is a homomorphism of rings.

Definition 1.2. Let R be a ring and $I \subset R$ an ideal, then we call R/I the quotient ring of R by I .

Example 1.1. Consider the ring $\mathbb{Z}$ and the subset $n\mathbb{Z} \subset \mathbb{Z}$. We know that $n\mathbb{Z}$ is an abelian subgroup, but we also have for any $m \in \mathbb{Z}$:

$$(1.6) \quad m[n\mathbb{Z}] = \{mnj \mid j \in \mathbb{Z}\} = \{n(mj) \mid j \in \mathbb{Z}\} \subset n\mathbb{Z}$$

so $n\mathbb{Z}$ is an ideal. The corresponding quotient ring is given by $\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}_n$.

Lemma 1.1. *Let $\phi : R \rightarrow S$ be a homomorphism of rings; then $\ker \phi$ is an ideal.*

Proof. We already know that $\ker \phi$ is an additive subgroup of R , so it remains to show that $a \ker \phi, \ker \phi a \subset \ker \phi$. But if $b \in \ker \phi$ then $\phi(ab) = \phi(a)\phi(b) = \phi(a) \cdot 0_S = 0_S$ and similarly $\phi(ba) = 0_S$, so $ab, ba \in \ker \phi$ or $a \ker \phi, \ker \phi a \subset \ker \phi$. ■

Example 1.2. Let $F \leq E$ and $\alpha \in E$. Since $\phi_\alpha : F[x] \rightarrow E$ is a ring homomorphism it follows that:

$$(1.7) \quad \ker \phi_\alpha = \{f(x) \in F[x] \mid f(\alpha) = 0\},$$

is an ideal. If $\alpha \in F$ then $f(\alpha) = 0$ if and only if $f(x) = (x - \alpha)q(x)$, so we see that:

$$(1.8) \quad \ker \phi_\alpha = (x - \alpha)F[x] = \{(x - \alpha)q(x) \mid q(x) \in F[x]\}.$$

As you will show in the homework $f(x) = (x - \alpha)q(x) + f(\alpha)$, so it follows that $\psi : F \rightarrow F[x]/\ker \phi_\alpha$ defined by $\psi(a) = a + \ker \phi_\alpha$ is surjective. It is also injective since a is never divisible by $x - \alpha$. So $F[x]/\ker \phi_\alpha$ is isomorphic as a ring to F .

Example 1.3. Let R be a commutative ring with unity and let $a \in R$ be any element. Then we define:

$$(1.9) \quad \langle a \rangle = \{ax \mid x \in R\}.$$

Notice that $ax + ay = a(x + y) \in \langle a \rangle$, $-ax = a(-x) \in \langle a \rangle$ and $0 = a0 \in \langle a \rangle$, so $\langle a \rangle$ is an additive subgroup of $(R, +)$. Moreover, if $y \in R$ then $yax = a(xy) \in \langle a \rangle$ so for any $ax \in \langle a \rangle$ and any $y \in R$ we have $yax \in \langle a \rangle$. That is, $\langle a \rangle$ is an ideal. This ideal is called the principal ideal generated by a .

Lemma 1.2. *Let $\phi : R \rightarrow S$ be a ring homomorphism, then:*

- (i) we have $\phi(0_R) = 0_S$,
- (ii) for $a \in R$ we have $\phi(-a) = -\phi(a)$,
- (iii) if $R' \leq R$ is a subring then $\phi(R')$ is a subring of S , item if $S' \leq S$ is a subring then $\phi^{-1}(S') \leq R$ is a subring,
- (iv) if $I \subset R$ is an ideal then $\phi(I) \subset \phi(R)$ is an ideal,
- (v) if $J \subset S$ is an ideal then $\phi^{-1}(J) \subset R$ is an ideal.

Remark 1. Notice that point (v) generalizes the fact that $\ker \phi$ is an ideal since $\{0_S\} \subset S$ is an ideal and $\ker \phi = \phi^{-1}(\{0_S\})$.

Proof. Notice that (i) and (ii) follows from the corresponding properties of groups: $\phi : (R, +) \rightarrow (S, +)$ is a homomorphism of abelian groups, so (i) and (ii) corresponds to the fact that group homomorphisms map identity to identity, and inverses to inverses.

We will leave the proof of (iii) and (iv) as an exercise, and prove (v). Let $J \subset S$ be an ideal, we need to show that if $a \in \phi^{-1}(J)$ and $b \in R$ then $ab, ba \in \phi^{-1}(J)$. We will prove that $ab \in \phi^{-1}(J)$, that $ba \in \phi^{-1}(J)$ is proved analogously. We calculate:

$$(1.10) \quad \phi(ab) = \phi(a)\phi(b)$$

but $\phi(a) \in J$ so, since J is an ideal, we have $\phi(a)\phi(b) \in J$. Since $\phi(ab) = \phi(a)\phi(b) \in J$ it follows that $ab \in \phi^{-1}(J)$ which is what we wanted to prove. ■

Theorem 1.1 (Fundamental homomorphism theorem for rings). *Let $\phi : R \rightarrow S$ be a homomorphism of rings with kernel $I = \ker \phi$. Then there exists a unique isomorphism $\mu : R/I \rightarrow \phi(R)$ that satisfies $\mu \circ \gamma = \phi$.*

Proof. Recall that we can define $\mu(x + I) = \phi(x) + I$, and we already know from the last course that this map is an isomorphism of the groups $(R/I, +)$ and $(\phi(R), +)$. So, it remains to show that μ respects multiplication. To see this:

$$\begin{aligned} \mu((x + I)(y + I)) &= \mu(xy + I) = \phi(xy) + I = \phi(x)\phi(y) + I = \\ &= (\phi(x) + I)(\phi(y) + I) = \mu(x + I)\mu(y + I), \end{aligned}$$

which finishes the proof. ■

Any ring R always have two ideals, $\{0\}$ is an ideal, and R is an ideal. We call these ideals the trivial and the improper ideal respectively.

Definition 1.3. *An ideal $I \subset R$ is non-trivial and proper if $I \neq \{0\}$ and $I \neq R$.*

Lemma 1.3. *If R is a ring with unity and $I \subset R$ is an ideal that contains a unit $u \in I$ then $I = R$.*

Proof. Let $a \in R$ be arbitrary. Then $u \cdot (a/u) \in I$ since $u \in I$ and I is an ideal. But $u \cdot (a/u) = a$ so then $a \in I$. Since a was arbitrary we conclude that $I = R$. ■

Theorem 1.2. *A field F contains no proper non-trivial ideals.*

Proof. Note that every $a \in F \setminus \{0\}$ is a unit, so any ideal is either trivial or contains a unit and is therefore, by the previous lemma, all of F . ■

Definition 1.4. *Let R be a ring. An ideal $M \subset R$ is a maximal ideal if M is a proper ideal that is not properly contained in a proper ideal.*

Example 1.4. If p is prime then $p\mathbb{Z}$ is a proper ideal of $\mathbb{Z}$. Indeed, any ideal of $\mathbb{Z}$ is of the form $n\mathbb{Z}$. Now, if $p\mathbb{Z} \subset n\mathbb{Z}$ then there must be some $m \in \mathbb{Z}$ such that $p = nm$ which implies that either $n = p$ and $m = 1$ or $n = 1$ and $m = p$. In the first case we would have $n\mathbb{Z} = p\mathbb{Z}$, so $p\mathbb{Z} \subset n\mathbb{Z}$ is not a proper inclusion, and in the second case we have $n\mathbb{Z} = \mathbb{Z}$ so $n\mathbb{Z}$ is not a proper ideal.

Theorem 1.3. *Let R be a commutative ring with unity. An ideal $M \subset R$ is a maximal ideal if and only if R/M is a field.*

Proof. Note that the ring homomorphism $\gamma : R \rightarrow R/M$ is surjective, so if $M \subset N$ with N an ideal then $\gamma(N) \subset R/M$ is an ideal in R/M . Now, if R/M is a field then either $\gamma(N)$ is trivial (which correspond to $N = M$) or $\gamma(N) = R/M$ (which corresponds to $N = R$).

It remains to show that if M is a maximal ideal then R/M is a field. Note that R/M is commutative (since R is) and has a unity $1+M$: $(1+M)(x+M) = (1 \cdot x) + M = x + M$. Let $x + M \in R/M$ be non-zero. Consider the ideal $\langle x \rangle + M$ (from homework this is an ideal). Since $x \notin M$ (if $x \in M$ then $x + M = 0 + M$ would be zero) it holds that M is properly contained in $\langle x \rangle + M$, so since M is maximal we conclude that $\langle x \rangle + M = R$. In particular, there must be some $a \in R$ such that $ax + m = 1$ but this implies that $1 + M = ax + M = (a + M)(x + M)$, so $x + M$ has a multiplicative inverse. Since any non-zero element in R/M has a multiplicative inverse we conclude that R/M is a field. ■

Corollary 1.3.1. *A commutative ring with unity R is a field if and only if R has no proper non-trivial ideals.*

Proof. One direction we have already shown. If R is a commutative ring with unity with no proper non-trivial ideals then $\{0\}$ is a maximal ideal so $R \cong R/\{0\}$ is a field. ■

Definition 1.5. *An ideal $P \subset R$ is a prime ideal if $ab \in P$ implies that either $a \in P$ or $b \in P$.*

Theorem 1.4. *Let R be a commutative ring with unity and $P \subset R$ an ideal. Then R/P is an integral domain if and only if P is a prime ideal.*

Proof. We already know that R/P is a commutative ring with unity, where the unity is given by $1+P$. If R/P is an integral domain and $ab \in P$ then $(a+P)(b+P) = 0+P$ which implies that either $a + P = P$ or $b + P = P$ which in turn implies $a \in P$ or $b \in P$.

Conversely, if P is a prime ideal and if $(a + P)(b + P) = 0 + P$ then $ab + P = P$ or $ab \in P$. Using that P is a prime ideal we conclude that $a \in P$ or $b \in P$ which implies $a + P = 0$ or $b + P = 0$. ■

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